EE115B *Digital Circuits*

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Combinational Logic Analysis

Digital Fundamentals
Chapter 5

Review

- Variable: the input
- Literal: a variable complemented or uncomplemented
- Implicant: a product term of literals obtained from the K-map such that when implicant = 1, F = 1
- Prime implicant (PI): The implicant which cannot be simplified further, i.e. no literal can be removed
- Essential prime implicant: The *prime implicants* which contain 1's that can only be grouped in 1 way
- Cover: A set of *prime implicants* which covers all 1's

CD \ AB	00	01	11	10
00	1	1	0	0
01	1	1	1	0
11	0	0	1	1
10	1	1	0	0

- Variable: A, B, C and D
- Literal: A, B′, etc
- Implicant: A′BC′, AB′CD, A′D′, etc
- Prime implicant (PI): A′C′, ACD, ABD, BC′D, A′D′, etc
- Essential prime implicant: ACD, not ABD, not BC′D, etc
- Cover: ACD, A′D′, A′C′ and (BC′D or ABD)

https://atozmath.com/KMap.aspx

Quine-McCluskey Method

Step 1: write down function in SOP form
Step 2: Group them, and iterate for a few iterations:

- Convert to minterms, than group according to number of 1s
- Reduce minterms by comparing adjacent groups.

Step 3: remove non-essential PIs by prime implicant table

https://atozmath.com/KMap.aspx

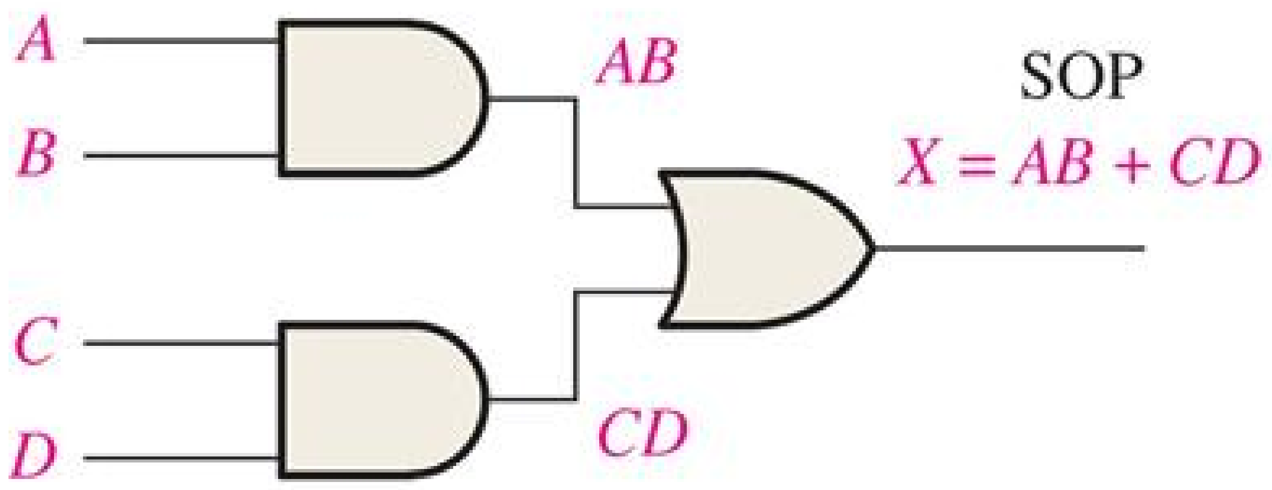
Basic Combinational Logic Circuits

Combinatorial Logic:
Refers to a type of digital logic circuit where the output is solely determined by the present input values, without any memory of past inputs.

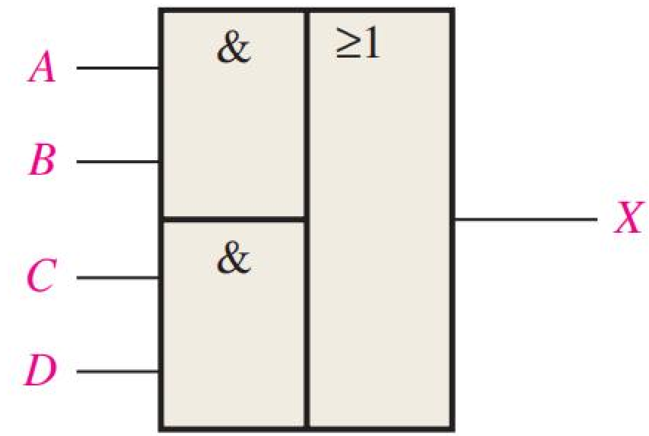
Previously, logic functions are represented as SOP and POS implementations.

So we will develop:
AND-OR circuits: for SOP
AND-OR-Invert circuit for POS

AND-OR Logic for SOP Expressions

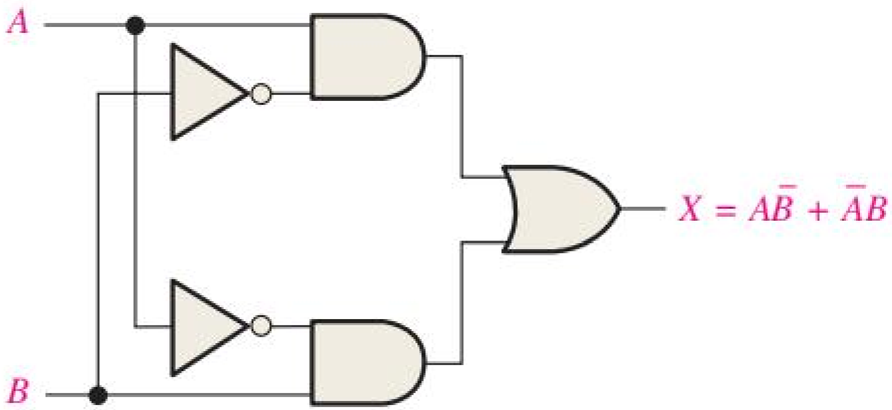


(a) Logic diagram (ANSI standard distinctive shape symbols)

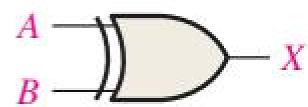


AND-OR Logic for SOP Expressions

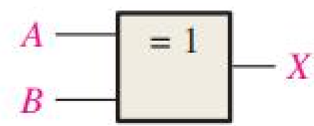
Exclusive-OR Logic



(a) Logic diagram

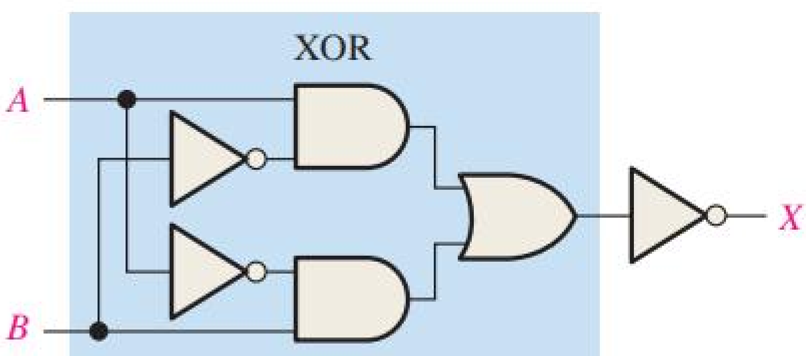


(b) ANSI distinctive shape symbol

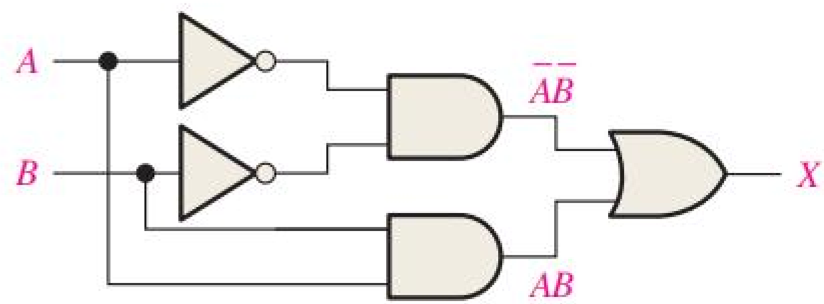


(c) ANSI rectangular outline symbol

Exclusive-NOR Logic

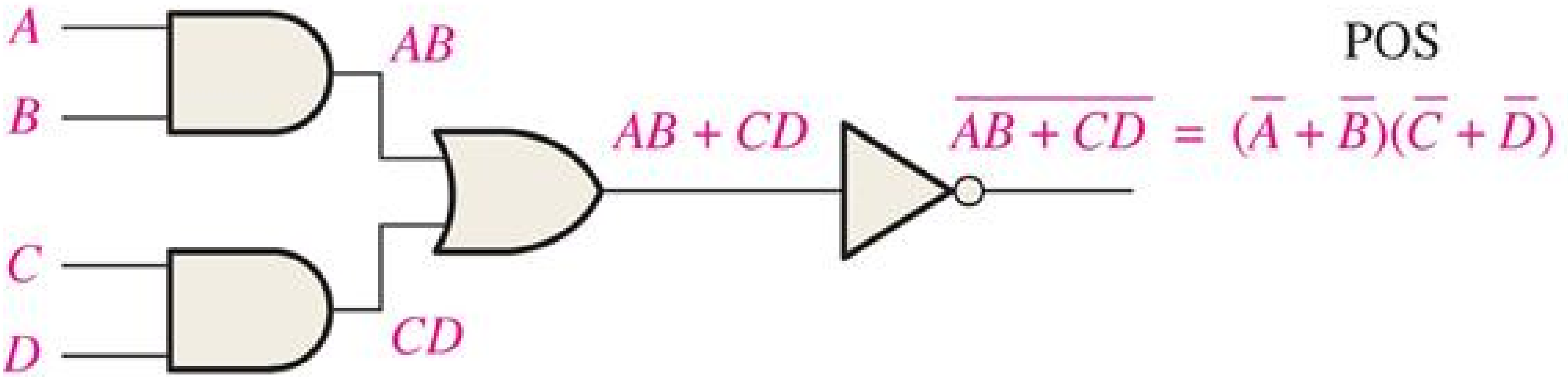


(a) $X = \overline{A\overline{B} + \overline{A}B}$

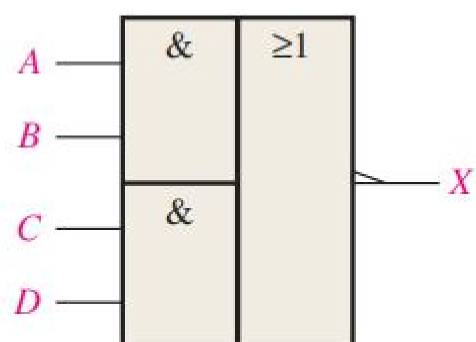


(b) $X = \overline{A}\overline{B} + AB$

AND-OR-Invert (AOI) Logic for POS Expressions



$$X = (\bar{A} + \bar{B})(\bar{C} + \bar{D}) = (\overline{AB})(\overline{CD}) = \overline{\overline{(\overline{AB})(\overline{CD})}} = \overline{\overline{\overline{AB}} + \overline{\overline{CD}}} = \overline{AB + CD}$$

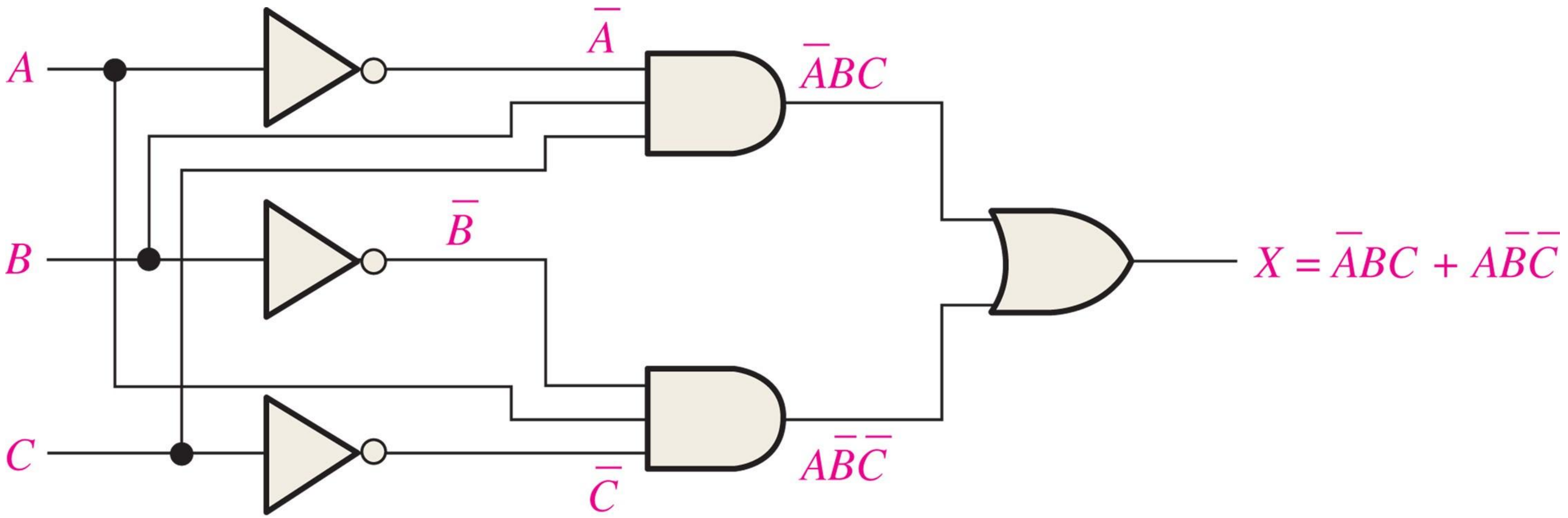


Exercise: implement this logic function as circuits

TABLE 5–3

Inputs			Output	
A	B	C	X	Product Term
0	0	0	0	
0	0	1	0	
0	1	0	0	
0	1	1	1	$\overline{A}BC$
1	0	0	1	$A\overline{B}\,\overline{C}$
1	0	1	0	
1	1	0	0	
1	1	1	0	

Exercise: implement this logic function as circuits



Universal Logic Element

NAND and NOR gates are widely known to be universal logic gates, meaning that any other logic gate be made from NAND or NOR gates

Advantages of Using Universal Gates

Simplify the design. Especially for FPGA or PLDs.

Why NAND/NOR?

AND/OR cannot invert logic.

XOR/XNOR cannot implement AND/OR.

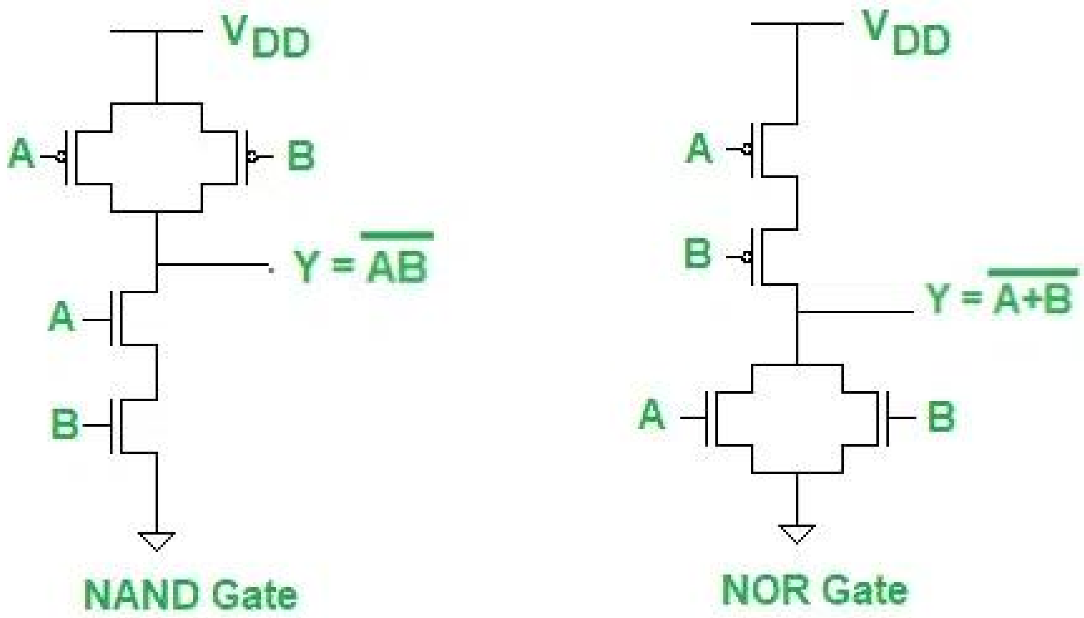
Disadvantages

Increased gate counts, higher power, and reduced speed.

https://www.geeksforgeeks.org/realization-of-logic-gate-using-universal-gates/

Universal Logic Element

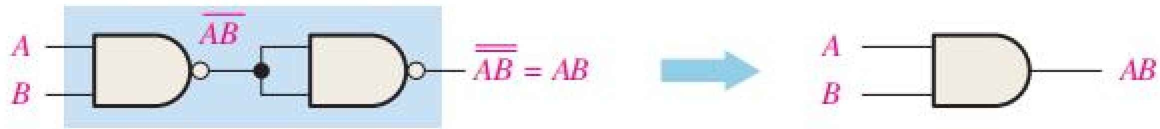
- CMOS implementations
  - 2-input AND/OR gate
  - 2-input NAND/NOR gate



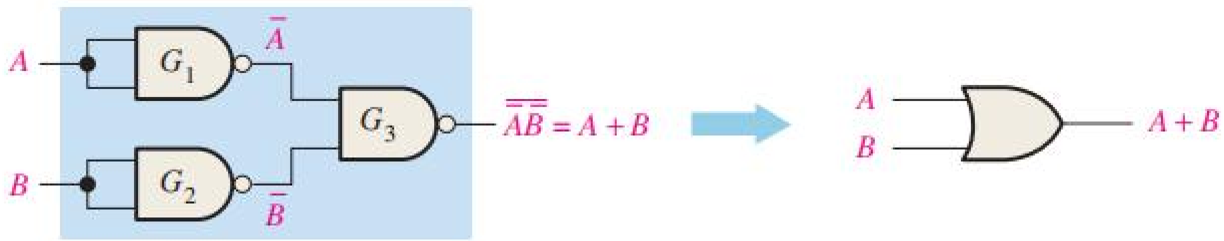
https://www.geeksforgeeks.org/cmos-logic-gate/

The NAND Gate as a Universal Logic Element

(a) One NAND gate used as an inverter



(b) Two NAND gates used as an AND gate

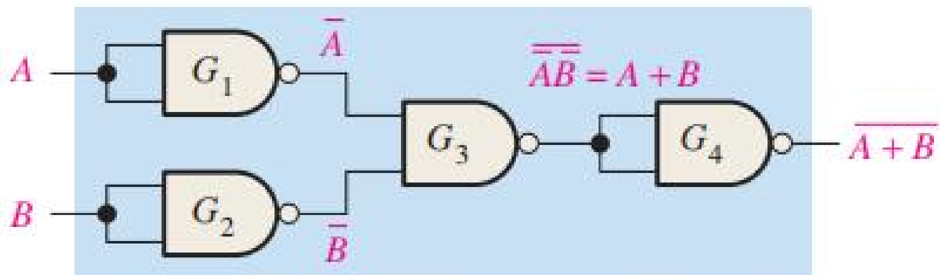


(c) Three NAND gates used as an OR gate

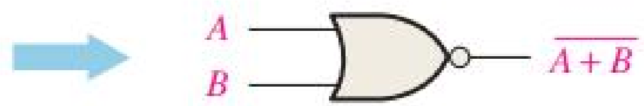
Exercise: Implement a NOR gate using NAND only?

The NAND Gate as a Universal Logic Element

Exercise: Implement a NOR gate using NAND?



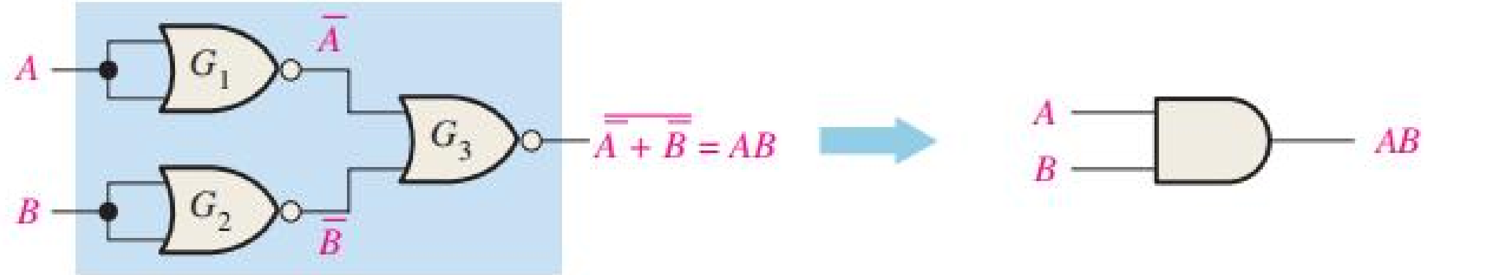
(d) Four NAND gates used as a NOR gate



The NOR Gate as a Universal Logic Element

(a) One NOR gate used as an inverter

(b) Two NOR gates used as an OR gate

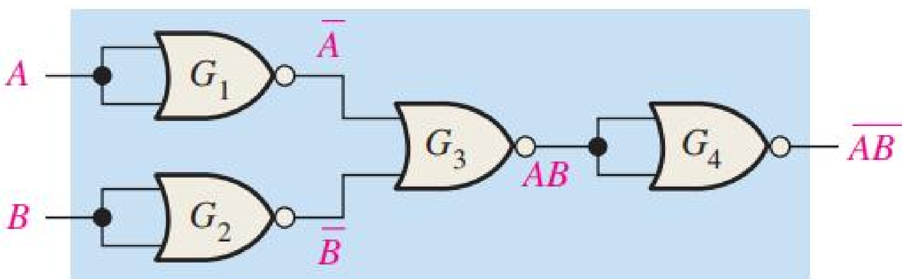


(c) Three NOR gates used as an AND gate

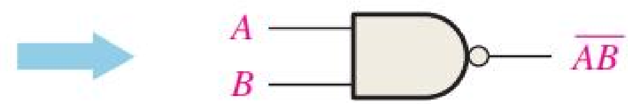
Exercise: Implement a NAND gate using NOR?

The NOR Gate as a Universal Logic Element

Exercise: Implement a NAND gate using NOR?



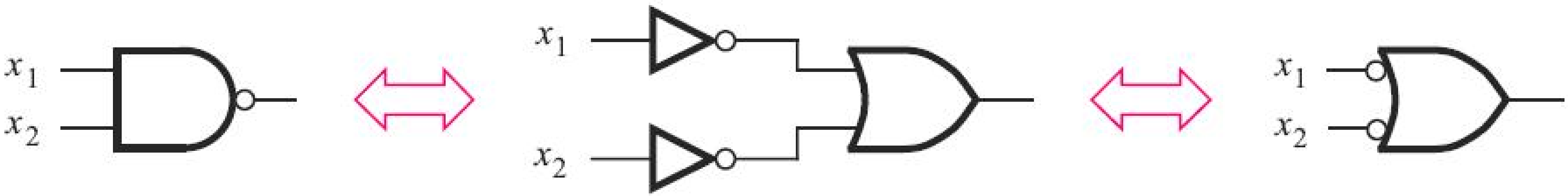
(d) Four NOR gates used as a NAND gate



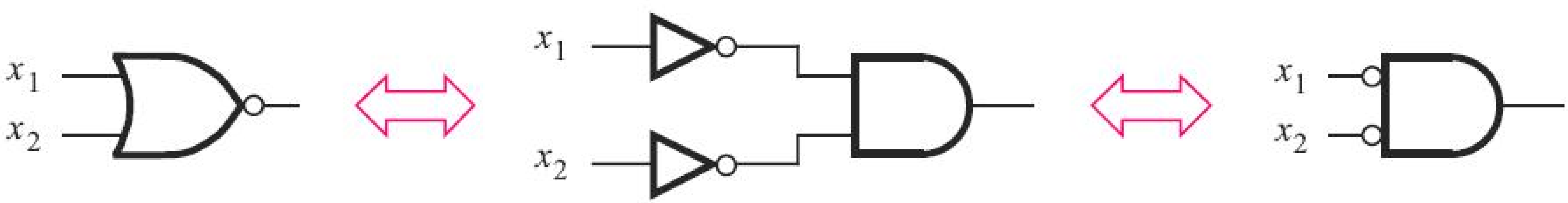
Universal Gate Element: Conversions

How can we implement any combinational circuit using NAND NOR?

Can we do conversion using a visualizable approach? (De Morgan's theorem)

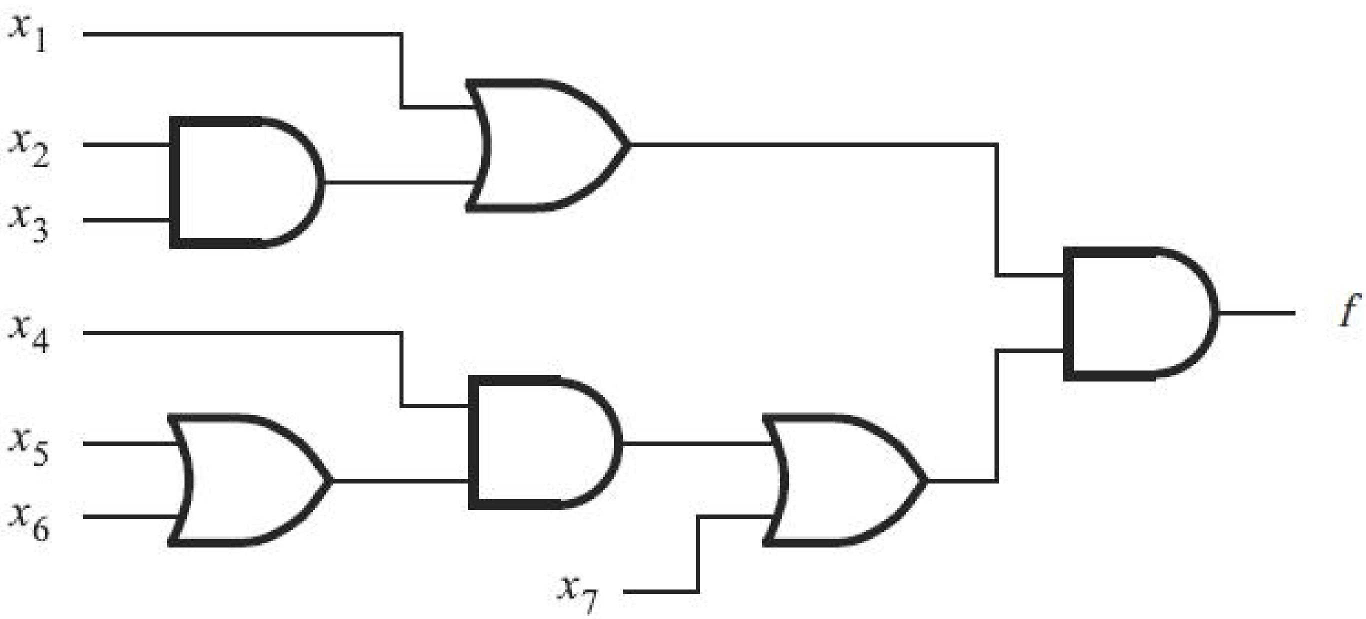


(a) $\overline{x_1 x_2} = \bar{x}_1 + \bar{x}_2$



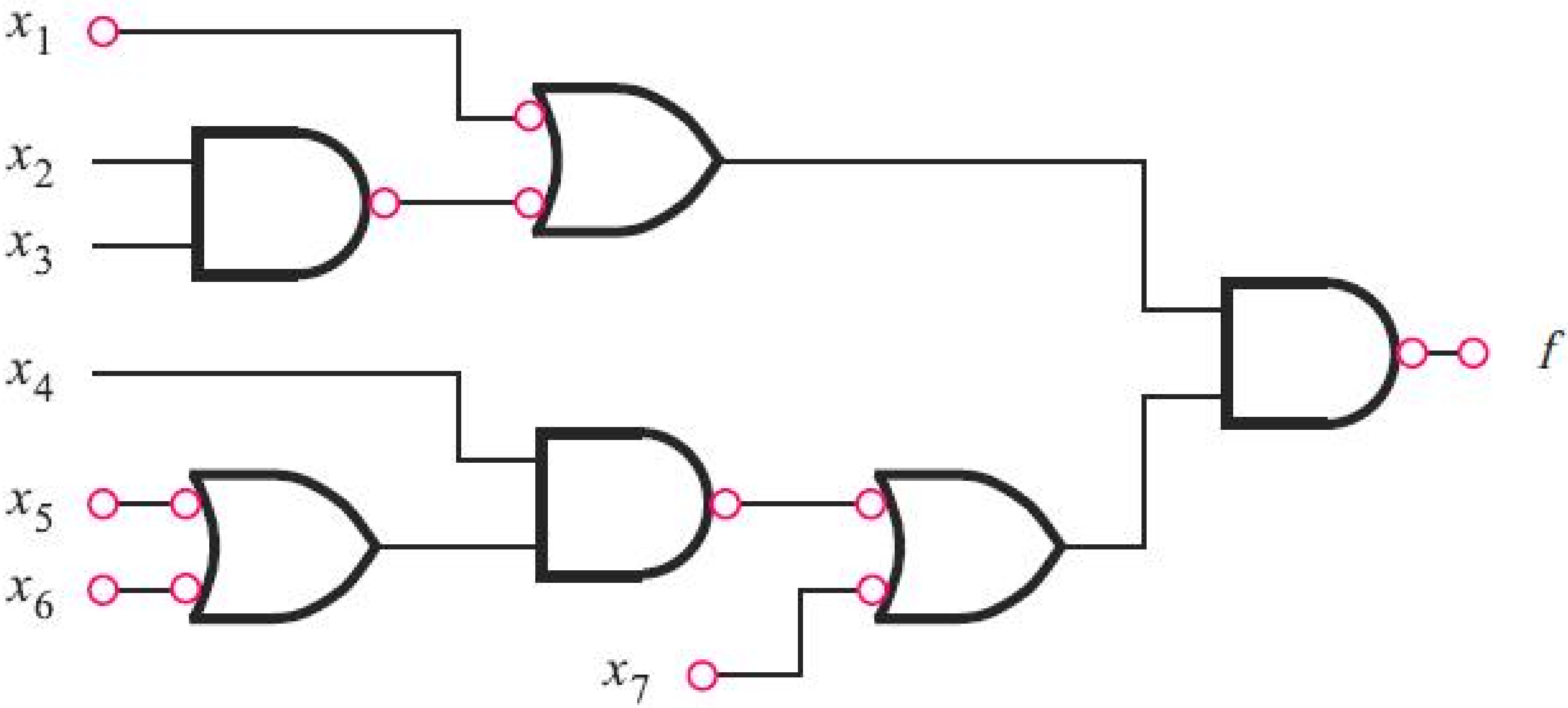
(b) $\overline{x_1 + x_2} = \bar{x}_1 \bar{x}_2$

Exercise 1: Convert to NAND only



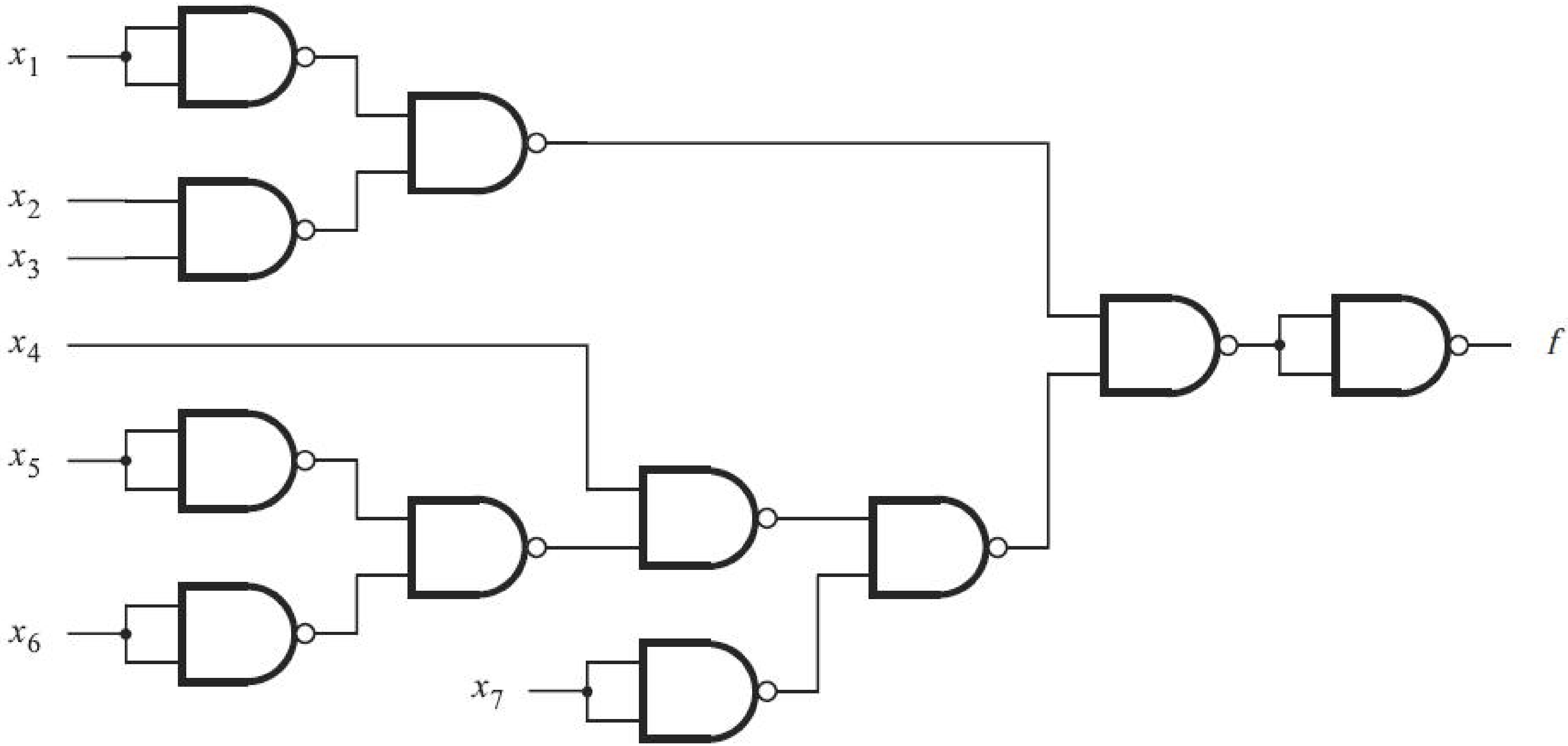
(a) Circuit with AND and OR gates

Exercise 1: Convert to NAND only



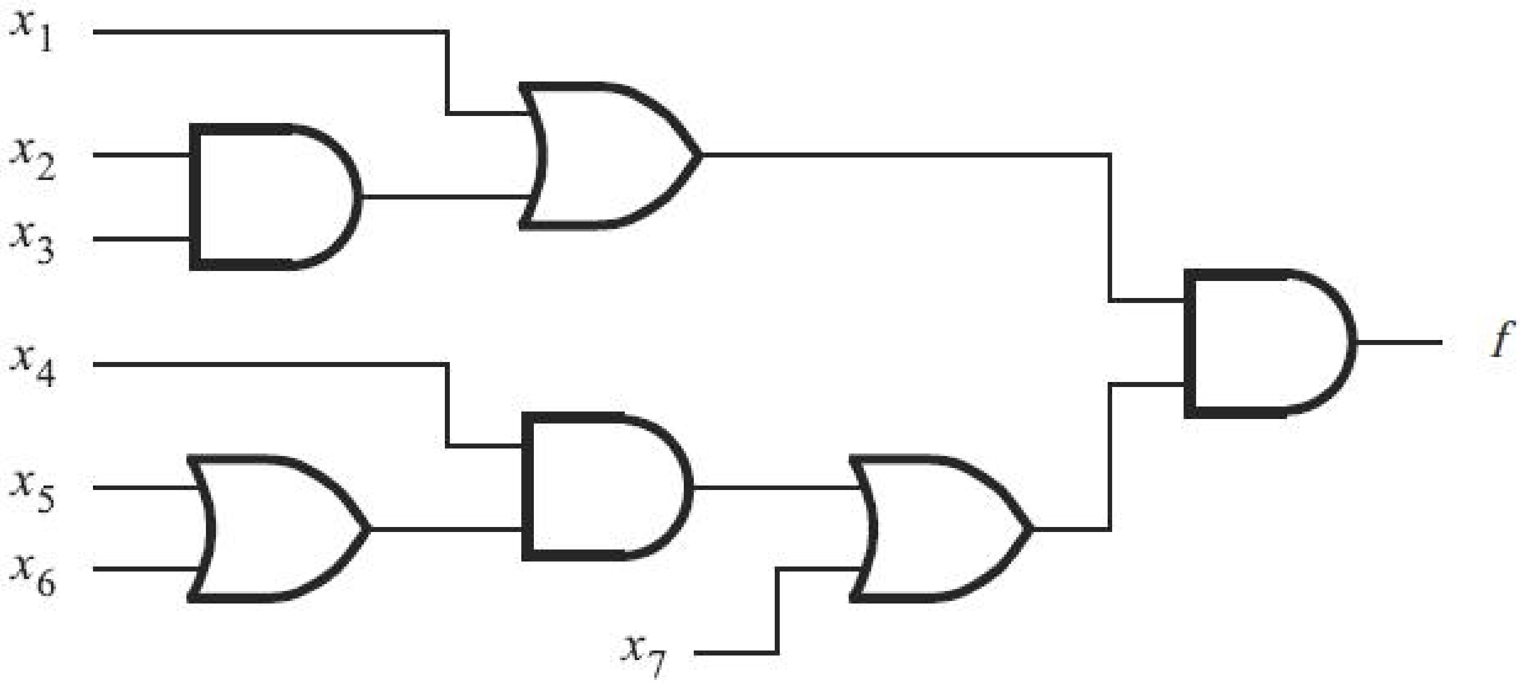
(b) Inversions needed to convert to NANDs

Exercise 1: Convert to NAND only



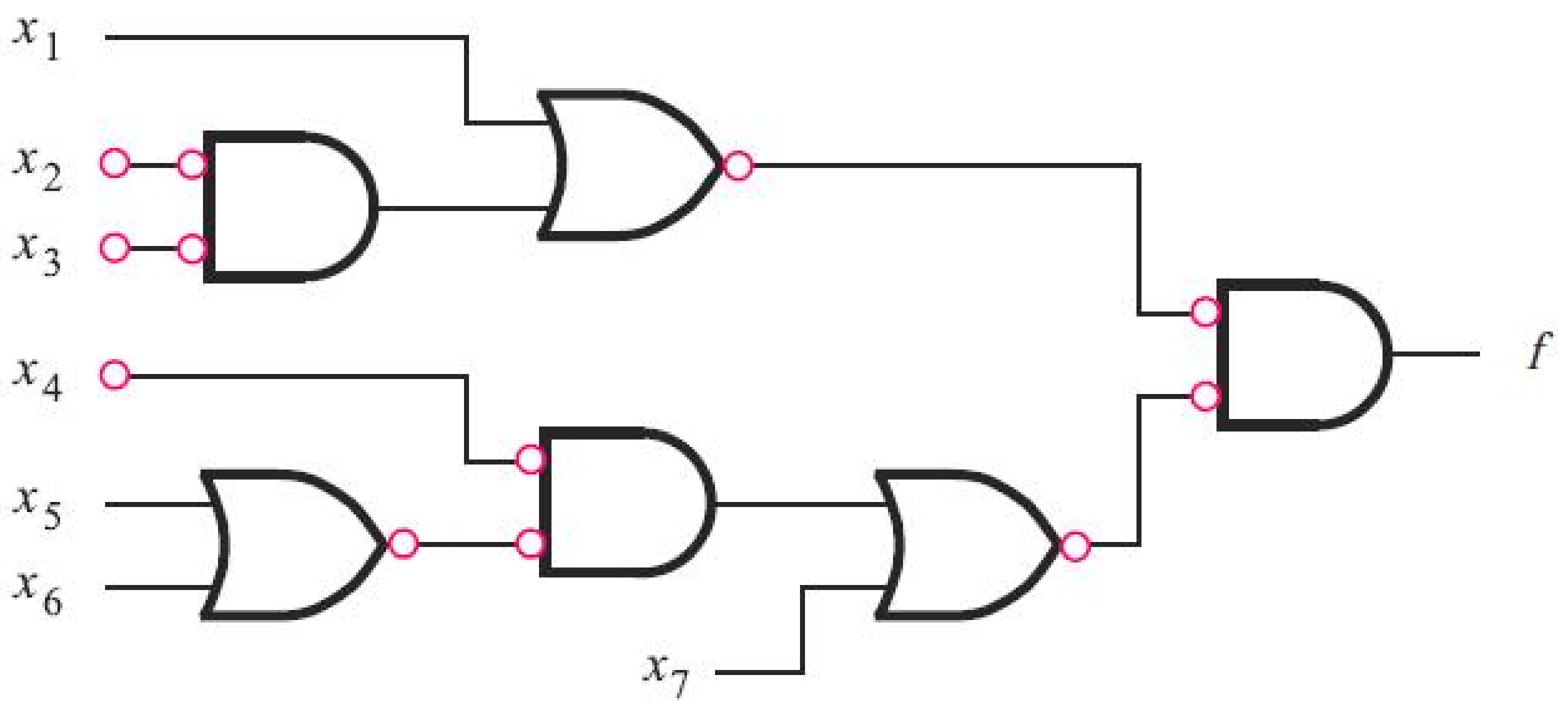
(c) NAND-gate circuit

Exercise 2: Convert to NOR only



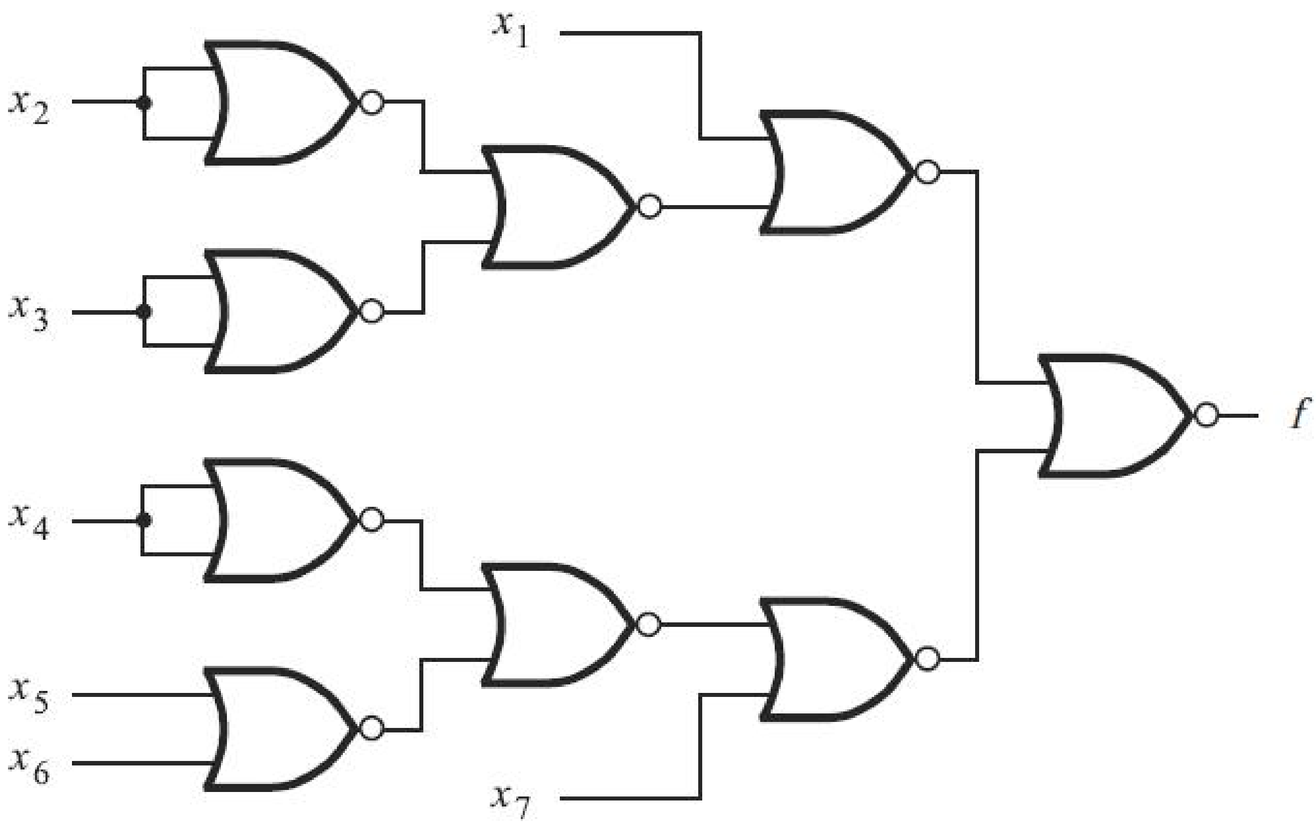
(a) Circuit with AND and OR gates

Exercise 2: Convert to NOR only



(a) Inversions needed to convert to NORs

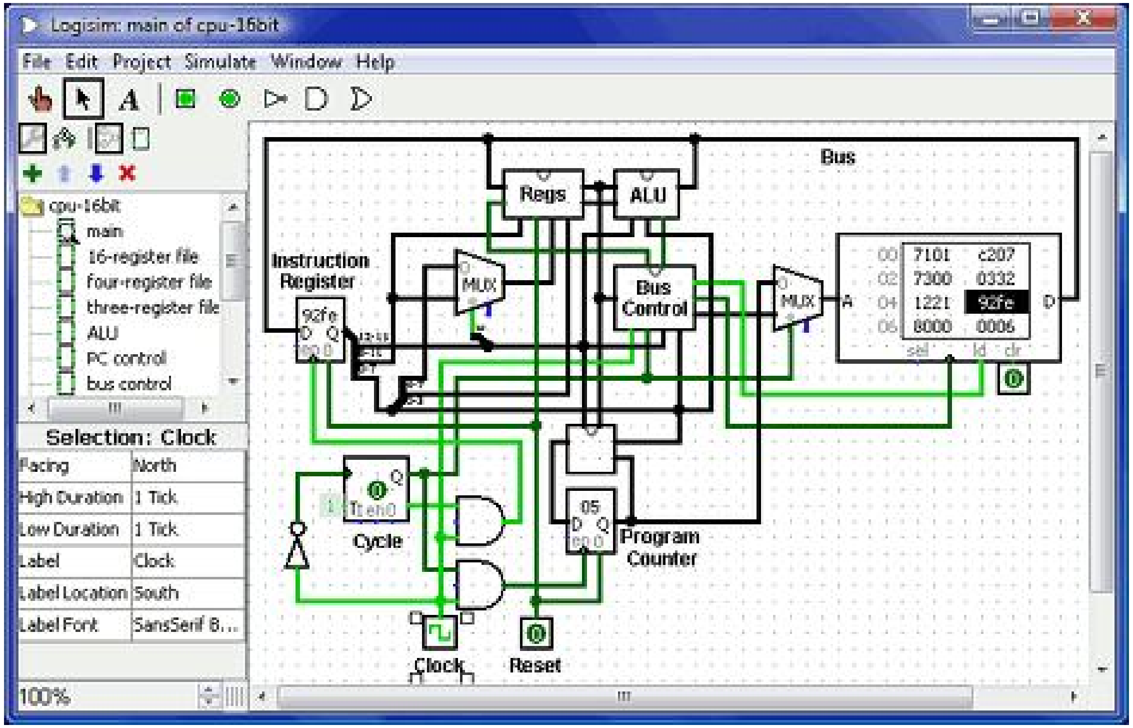
Exercise 2: Convert to NOR only



(b) NOR-gate circuit

Logisim Simulation

A free, open-source, graphical tool for logical simulation



https://www.cburch.com/logisim/

https://github.com/logisim-evolution/logisim-evolution

TI TINA Simulation

TINA-TI

SPICE-based analog simulation program

Downloads

Overview | Downloads | Technical documentation | Support & training

Overview

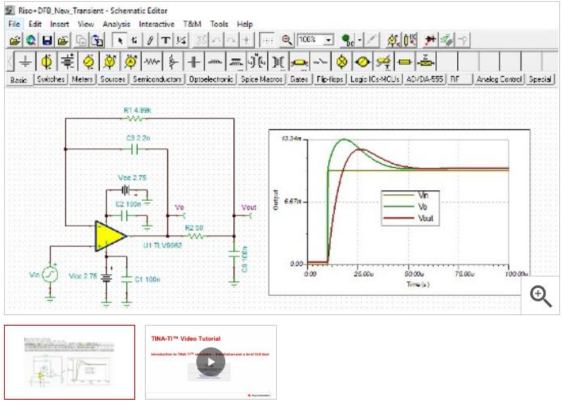
TINA-TI provides all the conventional DC, transient and frequency domain analysis of SPICE and much more. TINA has extensive post-processing capability that allows you to format results the way you want them. Virtual instruments allow you to select input waveforms and probe circuit nodes voltages and waveforms. TINA's schematic capture is truly intuitive - a real "quickstart."

TINA-TI installation requires approximately 500MB. Installation is straight-forward and it can be uninstalled easily, if you wish. We bet that you won't.

TINA is a product of DesignSoft exclusively for Texas Instruments. This complimentary version is fully functional but does not support some other features available with the full version of TINA.

For a complete list of available TINA-TI models, see: SpiceRack -- A Complete List

Need HSpice models to aid in your design? Our HSpice model collection can be found here.



https://www.ti.com.cn/tool/cn/TINA-TI

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